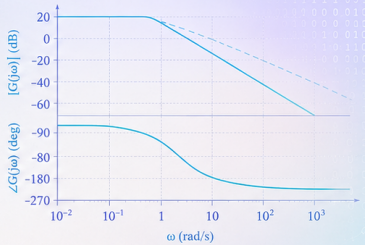
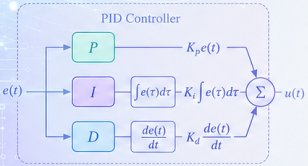
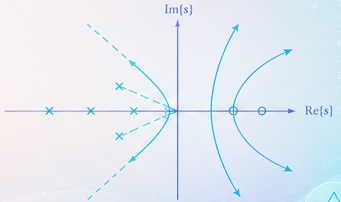


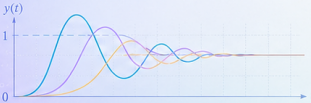
Stability Margin

Sahaj Saxena
sahajsaxena.github.io



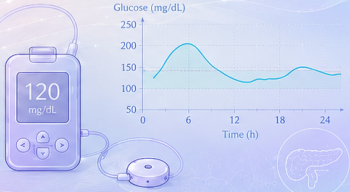






$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \cdots + b_m \frac{d^m u(t)}{dt^m} + \cdots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$





Absolute Stability versus Relative Stability

Main Question

A system may be stable, but how close is it to instability?

The Nyquist criterion tells us whether the closed-loop system is stable or unstable.

$$\boxed{Z = P - N}$$

However, two stable systems may have very different degrees of robustness.

Absolute Stability

- Is the closed-loop system stable?
- Are all closed-loop poles in the LHP?
- Nyquist answers this using encirclements.

Relative Stability

- How much gain increase can be tolerated?
- How much additional phase lag can be tolerated?
- Measured using gain margin and phase margin.

Gain Margin via the Nyquist Diagram

Let $L(s) = G(s)H(s)$. The phase-crossover frequency is defined by

$$\boxed{\angle L(j\omega_{pc}) = -180^\circ}.$$

If the Nyquist plot crosses the negative real axis with

$$|L(j\omega_{pc})| = r < 1,$$

then the gain may be increased by the factor

$$\boxed{GM = \frac{1}{r} = \frac{1}{|L(j\omega_{pc})|}}.$$

In decibels,

$$\boxed{GM_{\mathrm{dB}} = -20\log_{10}|L(j\omega_{pc})|.}$$

Gain Margin: Geometrical Interpretation

Suppose the Nyquist diagram crosses the negative real axis at

$$Q = -r, \qquad 0 < r < 1.$$

The critical point is

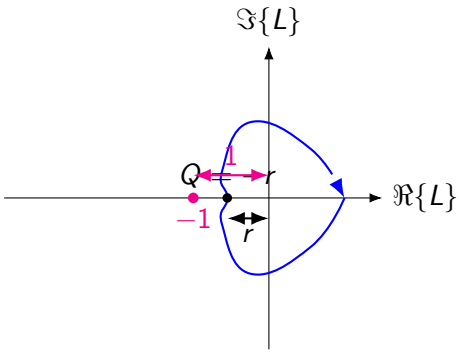
$$-1 + j0.$$

Thus,

$$GM = \frac{1}{r}$$

and

$$GM_{\mathrm{dB}} = 20 \log_{10} \frac{1}{r}.$$



Phase Margin via the Nyquist Diagram

The gain-crossover frequency is defined by

$$\boxed{|L(j\omega_{gc})| = 1}.$$

If

$$\angle L(j\omega_{gc}) = \phi_{gc},$$

then the additional phase lag required to reach -180° is

$$\boxed{PM = 180^\circ + \phi_{gc} = 180^\circ + \angle L(j\omega_{gc}).}$$

For example,

$$\phi_{gc} = -140^\circ \quad \Longrightarrow \quad \boxed{PM = 40^\circ}.$$

Phase Margin: Geometrical Interpretation

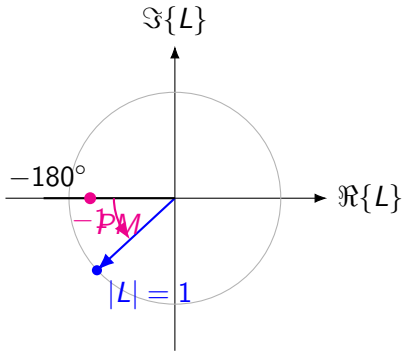
At the gain-crossover frequency,

$$|L(j\omega_{gc})| = 1.$$

The phase margin is the angular distance from the vector $L(j\omega_{gc})$ to the negative real axis.

$$\boxed{PM = 180^\circ + \angle L(j\omega_{gc})}$$

A larger positive phase margin generally indicates greater distance from instability.



Gain Margin and Phase Margin: Key Frequencies

Quantity	Frequency Condition	Margin
Gain margin	$\angle L(j\omega_{pc}) = -180^\circ$	$GM = \dfrac{1}{\lvert L(j\omega_{pc})\rvert}$
Phase margin	$\lvert L(j\omega_{gc})\rvert = 1$	$PM = 180^\circ + \angle L(j\omega_{gc})$

Terminology

ω_{pc} is the phase-crossover frequency; ω_{gc} is the gain-crossover frequency.

Gain and Phase Margins

Consider the loop transfer function

$$\boxed{L(s) = \frac{6}{(s^2 + 2s + 2)(s + 2)}}$$

For $s = j\omega$,

$$L(j\omega) = \frac{6\left[4(1 - \omega^2) - j\omega(6 - \omega^2)\right]}{16(1 - \omega^2)^2 + \omega^2(6 - \omega^2)^2}.$$

Objective

Determine the gain margin and phase margin.

Gain Margin

For

$$L(s) = \frac{6}{(s^2 + 2s + 2)(s + 2)},$$

the Nyquist plot crosses the negative real axis when

$$-\omega(6 - \omega^2) = 0.$$

Thus, for $\omega > 0$,

$$\boxed{\omega_{pc} = \sqrt{6} \text{ rad/s}}, \qquad L(j\sqrt{6}) = -0.3.$$

Therefore,

$$GM = \frac{1}{0.3} = 3.33, \qquad \boxed{GM_{\text{dB}} = 20 \log_{10}(3.33) = 10.45 \text{ dB}}.$$

Phase Margin

The gain-crossover frequency satisfies

$$|L(j\omega_{gc})| = 1.$$

Numerical evaluation gives

$$\boxed{\omega_{gc} \approx 1.253 \text{ rad/s}}, \qquad \angle L(j\omega_{gc}) \approx -112.3^\circ.$$

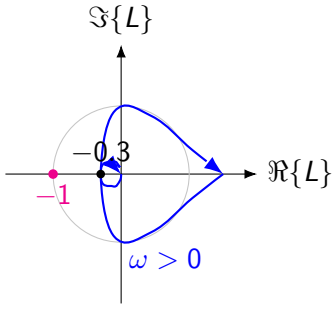
Hence,

$$\boxed{PM = 180^\circ - 112.3^\circ \approx 67.7^\circ.}$$

Interpretation

The positive margins show that the system is well separated from the stability boundary.

Nyquist Interpretation



$$GM = 3.33 = 10.45 \text{ dB}, \qquad PM = 67.7^\circ.$$

Stability via Bode Plots

Bode plots contain the same frequency-response information used by Nyquist.
For an open-loop stable system,

$$P = 0.$$

At the phase-crossover frequency,

$$\angle L(j\omega_{pc}) = -180^\circ.$$

Closed-loop stability requires

$$\boxed{|L(j\omega_{pc})| < 1 \quad \Longleftrightarrow \quad 20\log_{10}|L(j\omega_{pc})| < 0 \text{ dB}.}$$

Caution

For multiple crossovers or open-loop RHP poles, use the complete Nyquist criterion.

Gain and Phase Margins from Bode Plots

Gain Margin

Find the frequency where

$$\angle L(j\omega_{pc}) = -180^\circ.$$

Read the magnitude at the same frequency:

$$M_{pc} \text{ dB}.$$

Then

$$\boxed{GM_{\text{dB}} = -M_{pc}.}$$

Phase Margin

Find the frequency where

$$|L(j\omega_{gc})| = 1$$

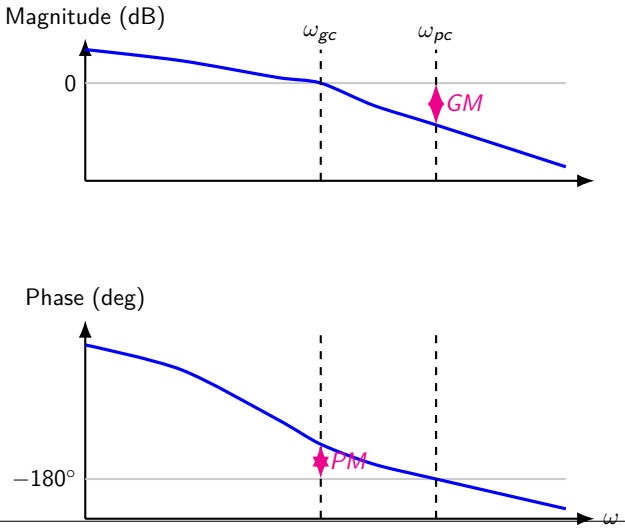
or

$$M(\omega_{gc}) = 0 \text{ dB}.$$

Then

$$\boxed{PM = 180^\circ + \phi(\omega_{gc}).}$$

Gain and Phase Margins on Bode Diagrams



Example 10.9: Range of Gain for Stability

Consider

$$\boxed{G(s) = \frac{K}{(s+2)(s+4)(s+5)}}.$$

All open-loop poles are in the LHP, so $P = 0$.
For convenience choose

$$K = 40.$$

Then

$$G(0) = \frac{40}{(2)(4)(5)} = 1 \quad \Longrightarrow \quad \boxed{20\log_{10}|G(0)| = 0 \text{ dB}}.$$

The break frequencies are

$$\boxed{2, \quad 4, \quad 5 \text{ rad/s}}.$$

Example: Bode Construction for $K = 40$

Magnitude

- $\omega < 2$: 0 dB/dec
- after 2: -20 dB/dec
- after 4: -40 dB/dec
- after 5: -60 dB/dec

At approximately

$$\omega \approx 7 \text{ rad/s},$$

the asymptotic magnitude is about

$$\boxed{-20 \text{ dB}}.$$

Phase

Each pole contributes approximately

$$0^\circ \to -90^\circ.$$

The combined phase reaches approximately

$$\boxed{-180^\circ}$$

at

$$\boxed{\omega \approx 7 \text{ rad/s}.}$$

Example: Range of K for Stability

From the asymptotic Bode plot for $K = 40$,

$$\omega_{pc} \approx 7 \text{ rad/s}, \qquad |G(j\omega_{pc})|_{\text{dB}} \approx -20 \text{ dB}.$$

Thus the gain may be increased by $+20$ dB before reaching the stability boundary.

$$10^{20/20} = 10, \qquad K_{cr} \approx 40(10) = 400.$$

Therefore,

$$\boxed{0 < K < 400 \quad \text{for stability.}}$$

Exact frequency-response analysis gives approximately

$$K_{cr} = 378, \qquad \omega \approx 6.16 \text{ rad/s}.$$

Example: Gain and Phase Margins for $K = 200$

For the same system,

$$G(s) = \frac{K}{(s+2)(s+4)(s+5)},$$

let

$$K = 200.$$

The previous Bode plot was drawn for $K = 40$. Therefore the magnitude shifts upward by

$$20 \log_{10}\left(\frac{200}{40}\right) = 20 \log_{10}(5) = \boxed{13.98 \text{ dB}}.$$

The phase plot is unchanged because multiplying by a positive constant gain does not change phase.

Example: Gain Margin

From the $K = 40$ Bode plot,

$$\angle G(j\omega) \approx -180^\circ$$

at approximately

$$\omega_{pc} \approx 7 \text{ rad/s}.$$

At this frequency the $K = 40$ magnitude is approximately

$$-20 \text{ dB}.$$

For $K = 200$, it becomes

$$-20 + 13.98 = -6.02 \text{ dB}.$$

Therefore,

$$\boxed{GM \approx 6.02 \text{ dB}.}$$

Example: Phase Margin

For $K = 200$, the actual magnitude plot is 13.98 dB above the $K = 40$ plot. Therefore the new 0 dB crossing is found where the original plot has magnitude

$$-13.98 \text{ dB}.$$

From the asymptotic Bode diagram,

$$\omega_{gc} \approx 5.5 \text{ rad/s}, \qquad \angle G(j\omega_{gc}) \approx -165^\circ.$$

Hence,

$$PM = 180^\circ - 165^\circ \approx 15^\circ.$$

Approximate Bode versus Exact Frequency Response

Hand-drawn Bode diagrams use straight-line magnitude and phase approximations. For the $K = 200$ example:

Quantity	Asymptotic Bode	Exact Response
Gain margin	$6.02\ \mathrm{dB}$	$\approx 5.53\ \mathrm{dB}$
Phase margin	15°	$\approx 23.5^\circ$
ω_{pc}	≈ 7	$\approx 6.16\ \mathrm{rad/s}$
ω_{gc}	≈ 5.5	$\approx 4.51\ \mathrm{rad/s}$

Teaching Point

Bode asymptotes are ideal for hand analysis; exact computational plots give more accurate margins.

MATLAB: Gain and Phase Margins

For

$$G(s) = \frac{200}{(s+2)(s+4)(s+5)},$$

the margins may be evaluated using

MATLAB

```
s = tf('s');
G = 200/((s+2)*(s+4)*(s+5));
margin(G);
grid on;
[Gm,Pm,Wpc,Wgc] = margin(G);
```

where

- *Gm*: gain margin as a ratio,
- *Pm*: phase margin in degrees,
- *Wpc*: phase-crossover frequency,
- *Wgc*: gain-crossover frequency.

Interpreting Stability Margins

Positive Margins

For the usual open-loop-stable case,

$$GM > 1 \quad (GM_{\mathrm{dB}} > 0)$$

and

$$PM > 0^\circ$$

indicate that the system has some distance from the stability boundary.

At the Stability Boundary

$$GM = 1 \quad (GM_{\mathrm{dB}} = 0)$$

or

$$PM = 0^\circ$$

means the Nyquist plot reaches

$$\boxed{-1 + j0}.$$

The closed-loop system is marginally stable.

Nyquist and Bode: Same Stability Information

Concept	Nyquist View	Bode View
Gain crossover	Unit-circle intersection	0 dB crossing
Phase crossover	Negative-real-axis crossing	-180° crossing
Gain margin	Radial distance to -1	dB distance to 0 dB
Phase margin	Angular distance to -180°	degree distance to -180°

Key Message

Nyquist provides the theoretical stability criterion; Bode plots provide a convenient engineering implementation for many practical systems.

Summary

Gain Margin

$$\boxed{GM = \frac{1}{|L(j\omega_{pc})|}}$$

$$\boxed{GM_{\mathrm{dB}} = -20\log_{10}|L(j\omega_{pc})|}$$

where

$$\angle L(j\omega_{pc}) = -180^\circ.$$

Phase Margin

$$\boxed{PM = 180^\circ + \angle L(j\omega_{gc})}$$

where

$$|L(j\omega_{gc})| = 1.$$

For open-loop-stable systems, positive margins indicate a useful distance from instability.

Thank You

Questions and Discussion